

Supplementary Information for

Toward rational understanding of the hydrogen evolution polarization curves through multiscale simulations

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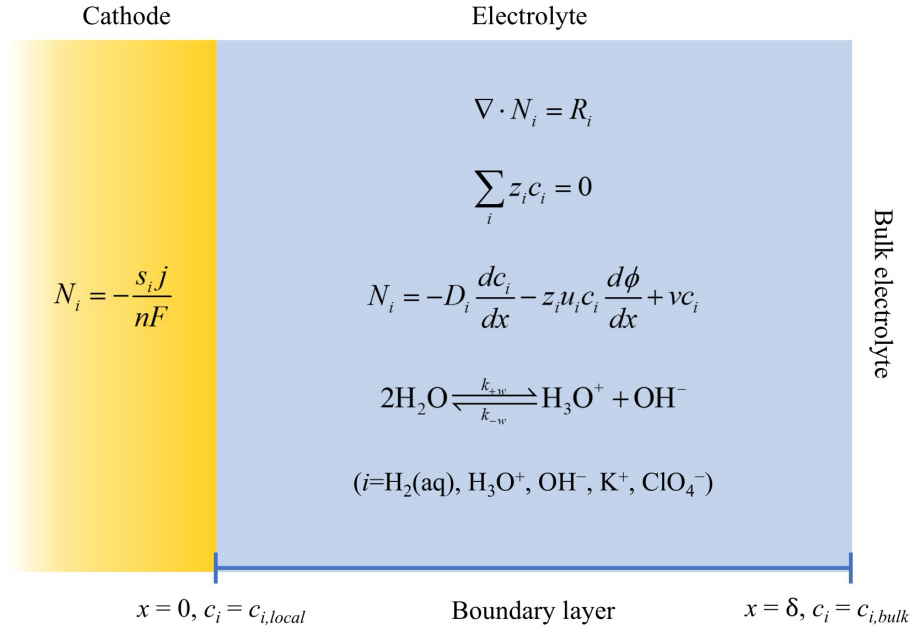
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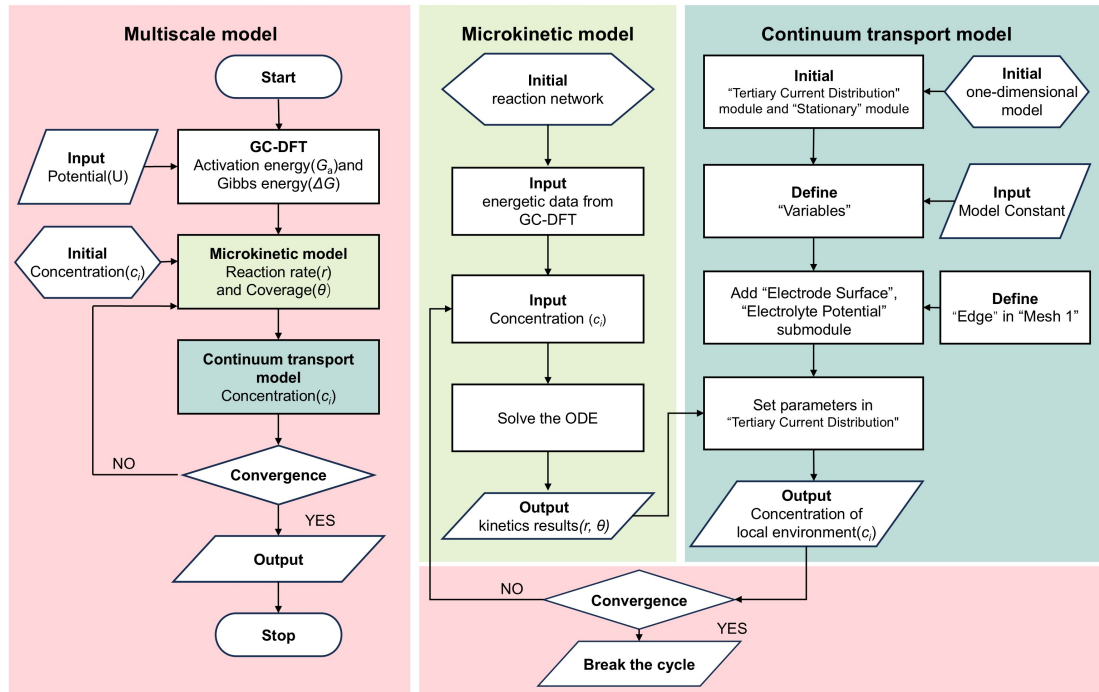
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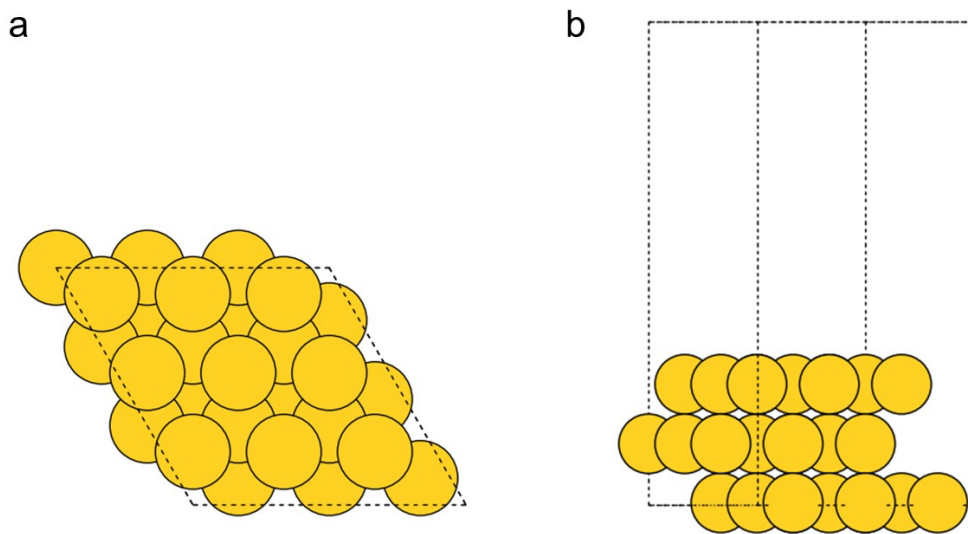
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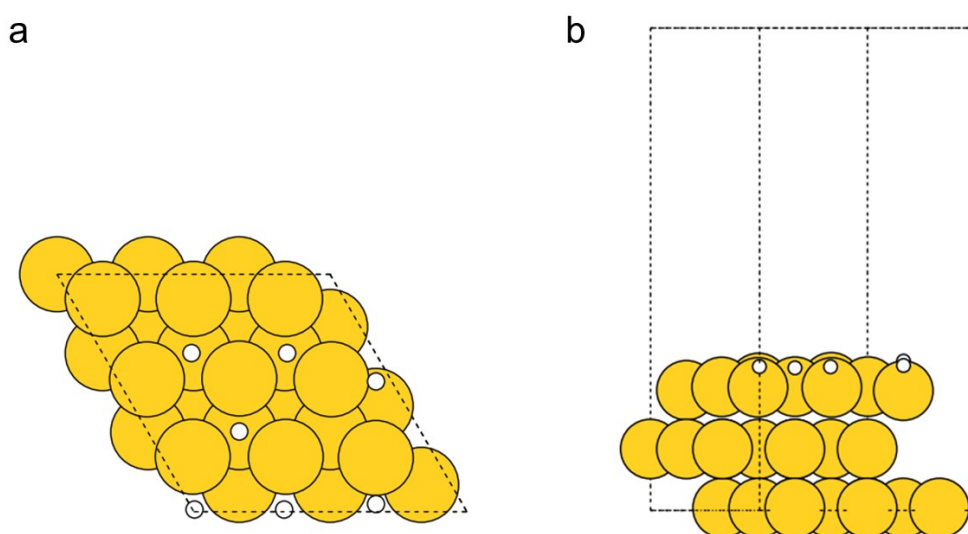
Supplementary Figure 1 | Governing equations and boundary conditions used for continuum transport model (where species i represents $\text{H}_2(\text{aq})$, H_2O , H_3O^+ , K^+ , and ClO_4^-).



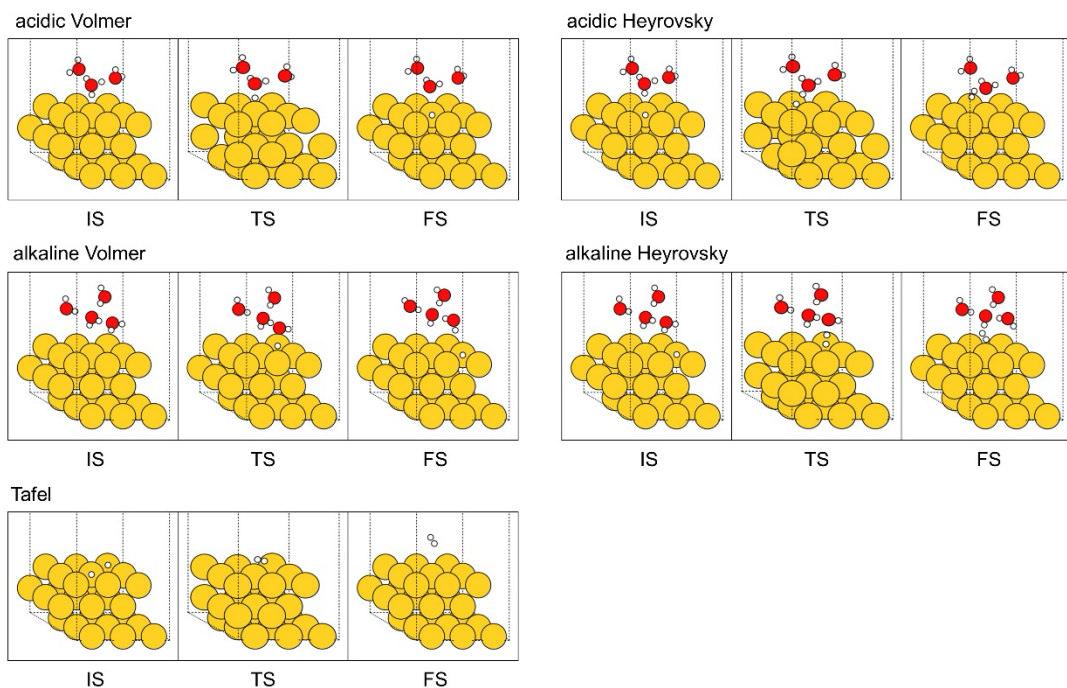
Supplementary Figure 2 | Multiscale simulation framework integrating continuum transport model, microkinetic model and GC-DFT.



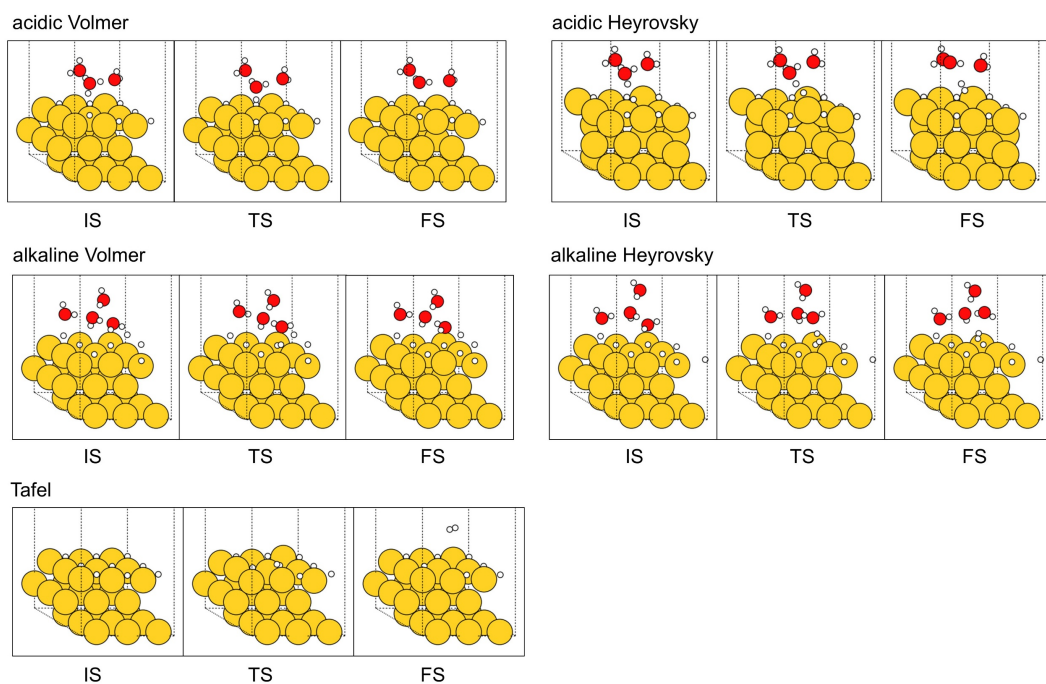
Supplementary Figure 3 | Top (a) and side (b) views of Au(111) surface model with θ_{*H} of 0 ML.



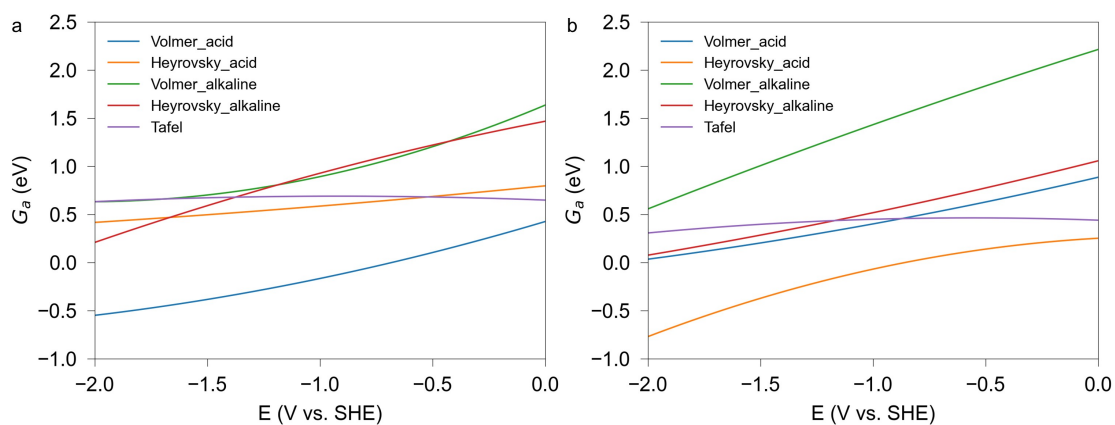
Supplementary Figure 4 | Top (a) and side (b) views of Au(111) surface model with θ_{*H} of 7/9 ML.



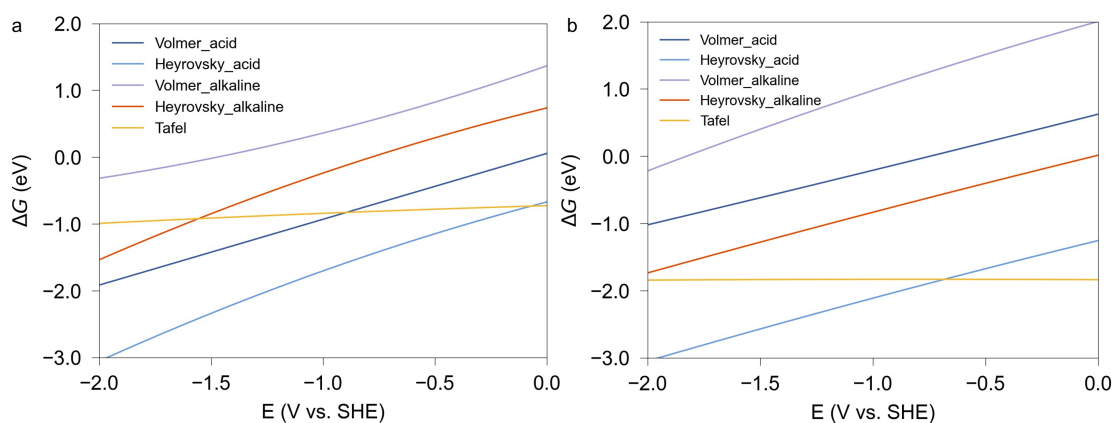
Supplementary Figure 5 | Optimized structures of initial (IS), transition (TS), and final (FS) states for all elementary steps on the Au(111) surface with θ_{*H} of 0 ML.



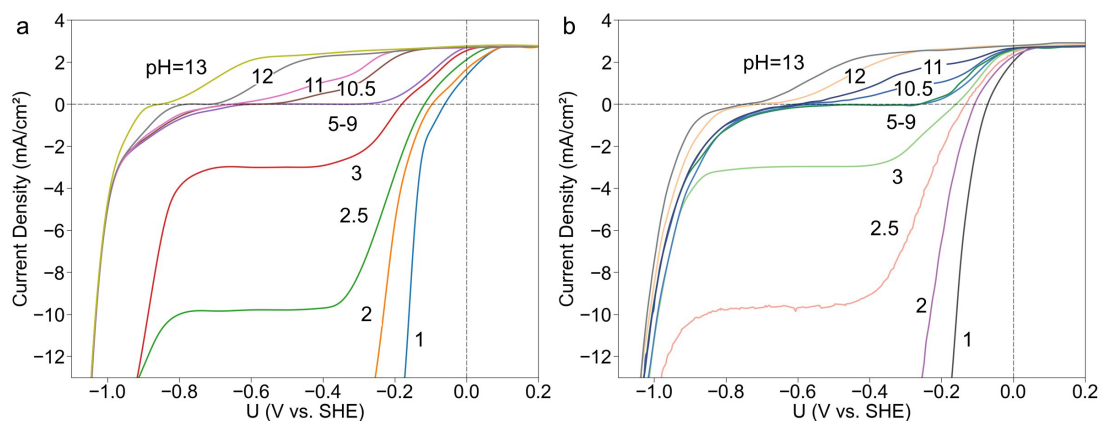
Supplementary Figure 6 | Optimized structures of initial (IS), transition (TS), and final (FS) states for all elementary steps on the Au(111) surface with θ_{*H} of 7/9 ML.



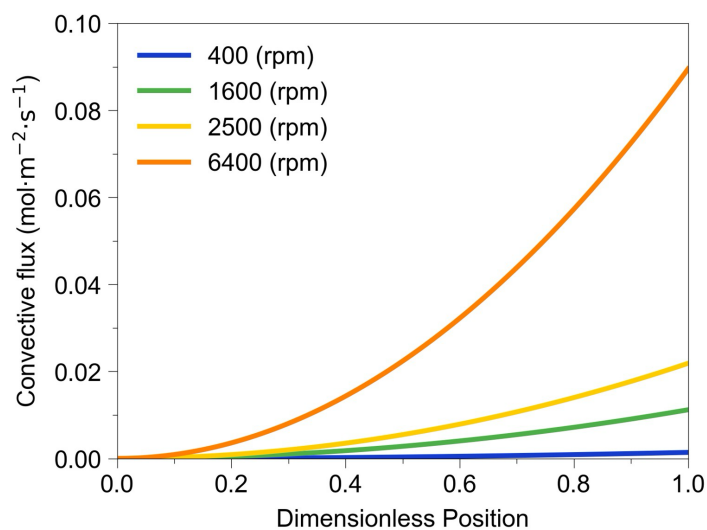
Supplementary Figure 7 | Calculated potential-dependent energy barriers (G_a) of all elementary steps on the Au(111) surface with θ^*_{H} of 0 ML (a) and 7/9 ML (b).



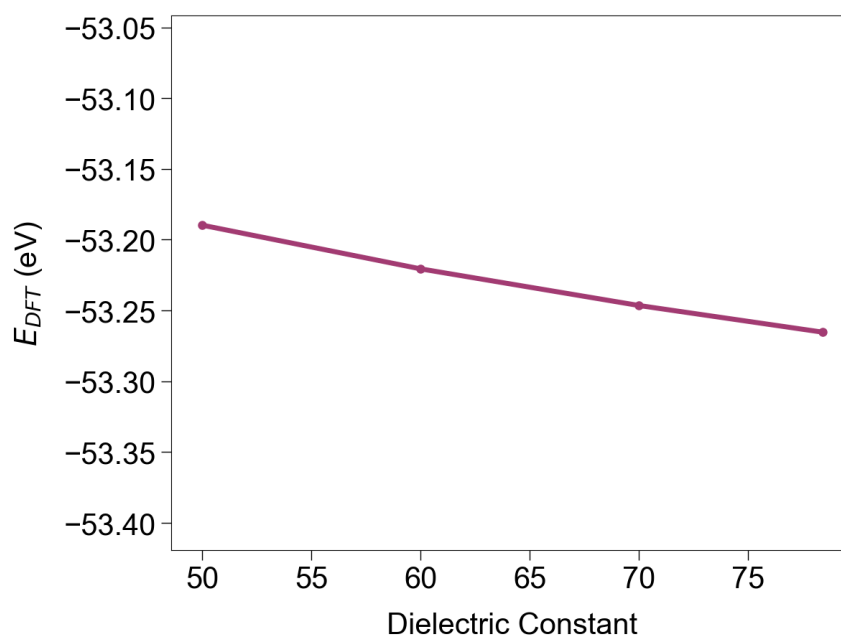
Supplementary Figure 8 | Calculated potential-dependent free energy change (ΔG) of all elementary steps on the Au(111) surface with θ^*_{H} of 0 ML (a) and 7/9 ML (b).



Supplementary Figure 9 | (a) Simulated current-potential polarization curves of HER on the Pt(111) surface. The original DFT calculation data were taken from literature.¹ (b) Experimental current-potential polarization curves on Pt(111) surface.² Copyright 2013 Springer Nature.



Supplementary Figure 10 | Convective flux within the boundary layer as a function of rotation speed.



Supplementary Figure 11 | Sensitivity analysis of dielectric constant effects on energetics. Energy profiles for the (H_8O_4) structure on Au(111) calculated with different relative permittivity ($\epsilon_r = 78.4, 70, 60, 50$) and their corresponding Debye lengths ($\lambda_D = 3.000, 3.175, 3.429, 3.757 \text{ \AA}$).

Supplementary Table 1. Diffusion coefficients of species in H₂O at 298.15 K within the model.

Species	Diffusion coefficient (10 ⁻⁹ m ² s ⁻¹)	Ref.
H ₃ O ⁺	8.000	1
OH ⁻	5.273	1
K ⁺	1.957	3
ClO ₄ ⁻	1.792	4
H ₂	3.700	1

Supplementary Table 2. List of fixed model parameters

Parameter	value	unit
Temperature(T)	298.15	K
Universal gas constant(R)	8.314	J mol ⁻¹ K ⁻¹
Faraday's constant(F)	96485	C mol ⁻¹
Viscosity(ν)	1.009×10 ⁻⁶	m ² s ⁻¹
Avogadro constant(N_A)	6.0221×10 ²³	mol ⁻¹
Boundary layer thickness(δ)	4×10 ⁻⁴	m
Bulk concentration of H ₂ O	55.6	mol L ⁻¹
Bulk concentration of H ₃ O ⁺	10 ^{-pH}	mol L ⁻¹
Bulk concentration of OH ⁻	10 ^{pH-14}	mol L ⁻¹
Bulk concentration of H ₂	0.0007	mol L ⁻¹
Bulk concentration of K ⁺	0.1+10 ^{pH-14}	mol L ⁻¹
Bulk concentration of ClO ₄ ⁻	0.1+10 ^{-pH}	mol L ⁻¹
concentration of active sites (c^*)	2.16×10 ⁻⁵	mol m ⁻²

Supplementary Table 3. The calculated work function (W_f), potentials (U) and DFT energies (E_{DFT}) of all species on the Au(111) surface with θ_{H} of 0 ML.

Species	Δq (e)	E_{elect} (e)	W_f (eV)	U (V vs. SHE)	E_{DFT} (eV)
H_3O^{++*}	0	322	4.5367	0.2067	-43.9615
	0.5	322.5	4.0081	-0.3219	-46.9010
	1	323	3.4358	-0.8942	-49.5747
	1.5	323.5	2.7492	-1.5808	-51.9244
TS_1	0.8	322.8	4.1343	-0.1957	-48.1685
	1	323	3.8403	-0.4897	-49.2897
	1.3	323.3	3.3671	-0.9629	-50.8548
	1.8	323.8	2.4971	-1.8329	-53.1301
$\text{H}_2\text{O}+\text{H}^*$	0.5	322.5	4.7894	0.4594	-46.5817
	1	323	4.2571	-0.0729	-49.6525
	1.5	323.5	3.6656	-0.6644	-52.4694
	2.5	324.5	2.4769	-1.8531	-57.2089
$\text{H}_3\text{O}^++\text{H}^*$	0	323	4.5093	0.1793	-47.1745
	0.5	323.5	3.9035	-0.4265	-50.0833
	1.5	324.5	2.6342	-1.6958	-54.9573
	2	325	2.0994	-2.2306	-56.9361
TS_2	0.3	323.3	4.3937	0.0637	-48.1168
	0.8	323.8	3.7893	-0.5407	-50.9560
	1.3	324.3	3.1305	-1.1995	-53.4881
	1.5	324.5	2.8404	-1.4896	-54.4151
$\text{H}_2\text{O}+\text{H}_2(\text{aq})$	0.5	323.5	4.5775	0.2475	-50.4401
	1	324	4.2697	-0.0603	-53.6116
	1.5	324.5	3.7293	-0.6007	-56.4332
	2	325	3.2549	-1.0751	-59.1494
$\text{H}_2\text{O}+^*$	1	330	4.3677	0.0377	-59.7325
	1.5	330.5	3.8284	-0.5016	-62.5956
	2	331	3.2114	-1.1186	-65.1670
	2.5	331.5	2.615	-1.715	-67.5341
TS_3	1.3	330.3	4.7719	0.4419	-59.6540
	1.8	330.8	4.1583	-0.1717	-62.6759
	2.3	331.3	3.4585	-0.8715	-65.3935
	3.0	332	2.4716	-1.8584	-68.5883
OH^-+H^*	1.5	330.5	4.7438	0.4138	-61.0671
	2	331	4.1966	-0.1334	-64.1037

	2.5	331.5	3.584	-0.746	-66.8721
	3	332	2.9179	-1.4121	-69.3326
H ₂ O+H*	1	331	4.3532	0.0232	-63.0046
	1.5	331.5	3.7247	-0.6053	-65.8305
	2	332	3.0597	-1.2703	-68.3311
	2.5	332.5	2.485	-1.8450	-70.5802
TS_4	1.8	331.8	4.0554	-0.2746	-66.1887
	2	332	3.8043	-0.5257	-67.2980
	2.3	332.3	3.411	-0.919	-68.8664
	2.5	332.5	3.1453	-1.1847	-69.8482
OH+H ₂ (aq)	1.5	330.5	3.8284	-0.5116	-54.4680
	2	331	3.2114	-1.1286	-55.5641
	2.5	331.5	2.615	-1.7250	-57.0216
	3	332	1.8782	-2.4618	-59.0391
2H*	1	300	4.2556	-0.0744	-13.0704
	1.5	300.5	3.6477	-0.6823	-15.8416
	2	301	3.000	-1.3300	-18.2940
	2.5	301.5	2.3796	-1.9504	-20.4099
TS_5	1	300	4.1745	-0.1555	-12.4650
	1.5	300.5	3.6118	-0.7182	-15.1929
	2	301	3.0426	-1.2874	-17.6398
	2.5	301.5	2.4479	-1.8821	-19.8020
H ₂ (aq)	1	300	4.332	0.0020	-13.8025
	1.5	300.5	3.7922	-0.5378	-16.6238
	2	301	3.1873	-1.1427	-19.1789
	2.5	301.5	2.4735	-1.8565	-21.3921

Supplementary Table 4. The calculated work function (W_f), potentials (U) and DFT energies (E_{DFT}) of all species on the Au(111) surface with θ_{H} of 7/9 ML.

Species	Δq (e)	E_{elect} (e)	W_f (eV)	U (V vs. SHE)	E_{DFT} (eV)
H_3O^{++*}	0	329	4.0759	-0.3541	-64.9814
	0.5	329.5	3.4446	-0.9854	-68.5904
	1	330	2.8083	-1.6217	-70.9584
	1.5	330.5	2.202	-2.228	-73.0197
TS_1	0	329	4.6332	0.2032	-64.9814
	0.5	329.5	4.0463	-0.3837	-67.9691
	1	330	3.3327	-1.0973	-70.6294
	1.5	330.5	2.5207	-1.9093	-72.8884
$\text{H}_2\text{O}+\text{H}^*$	0.5	329.5	4.3488	-0.0812	-68.1908
	1	330	3.8122	-0.6178	-71.0565
	1.5	330.5	3.2247	-1.2053	-73.6313
	2	331	2.6248	-1.8052	-75.9112
$\text{H}_3\text{O}^++\text{H}^*$	0.1	330.1	4.0409	-0.3891	-68.5684
	0.5	330.5	3.4249	-1.0051	-71.2288
	1.5	331.5	2.8200	-1.6100	-76.5870
	2	332	2.3316	-2.0984	-78.5918
TS_2	0	330	4.2329	-0.1971	-68.9208
	0.3	330.3	3.9151	-0.5149	-70.0490
	0.8	330.8	3.4199	-1.0101	-72.7275
	1	331	3.2886	-1.1414	-73.8067
$\text{H}_2\text{O}+\text{H}_2(\text{aq})$	0.5	330.5	4.3019	-0.1281	-72.7097
	1	331	3.8464	-0.5836	-75.5999
	1.5	331.5	3.2001	-1.2299	-78.1704
	2	332	2.6163	-1.8137	-80.4780
H_2O^{+*}	1	337	3.9647	-0.4653	-81.6115
	1.5	337.5	3.2974	-1.1326	-84.2221
	2	338	2.6383	-1.7917	-86.4863
	2.5	338.5	1.9817	-2.4483	-88.4392
TS_3	1	337	4.7151	0.2851	-79.2756
	1.5	337.5	4.1682	-0.2618	-82.3679
	2	338	3.4537	-0.9763	-85.0327
	2.5	338.5	2.7806	-1.6494	-87.3671
OH^-+H^*	1.5	337.5	4.3258	-0.1042	-82.5429
	2	338	3.8191	-0.6109	-85.3929

	2.5	338.5	3.2238	-1.2062	-87.9808
	3	339	2.5703	-1.8597	-90.2252
H ₂ O+H*	1	338	3.9319	-0.4981	-84.2502
	1.5	338.5	3.3228	-1.1072	-86.8737
	2	339	2.6738	-1.7562	-89.1720
	2.5	339.5	2.0255	-2.4045	-91.1568
TS_4	0.5	337.5	5.1844	0.7544	-80.0792
	1	338	4.5965	0.1665	-83.3374
	1.3	338.3	4.1886	-0.2414	-85.1200
	1.5	338.5	3.9525	-0.4775	-86.2544
OH ⁻ +H ₂ (aq)	1.5	338.5	4.3053	-0.1247	-87.1375
	2	339	3.7948	-0.6352	-89.9748
	2.5	339.5	3.143	-1.2870	-92.5348
	3	334	2.4592	-1.9708	-94.7293
2H*	0	306	4.8012	0.4712	-27.7696
	1	307	3.8809	-0.4491	-28.2641
	2	308	2.6417	-1.6883	-30.1393
	3	309	1.3953	-2.9347	-33.2357
TS_5	0.5	306.5	4.3419	0.0119	-27.4899
	1	307	3.8465	-0.4835	-27.8560
	1.5	307.5	3.3103	-1.0197	-28.5408
	2	308	2.6958	-1.6342	-29.6298
H ₂ (aq)	0	306	4.9261	0.5961	-29.5869
	1	307	3.934	-0.396	-30.0830
	2	308	2.6832	-1.6468	-31.8653
	3	309	1.4133	-2.9167	-35.0634

Supplementary Table 5. The potential-dependent free energy expressions ($G(U)$) of each species on the Au(111) surface with θ_{H} of 0 ML and 7/9 ML.

Species	$G(U)$ (eV) (0 ML)	$G(U)$ (eV) (7/9 ML)
H_3O^{++*}	$-0.4278U^2-0.0124U-43.1598$	$-0.3982U^2-0.2866U-65.1681$
TS_1	$-0.3232U^2+0.6839U-42.7352$	$-0.3392U^2+0.2572U-64.2835$
$\text{H}_2\text{O}+\text{H}^*$	$-0.4271U^2+0.9743U-43.1047$	$-0.3864U^2+0.5599U-64.5435$
$\text{H}_3\text{O}^++\text{H}^*$	$-0.4123U^2+0.1420U-46.3088$	$-0.4364U^2-0.3614U-67.7368$
TS_2	$-0.3917U^2+0.3732U-45.5135$	$-0.6265U^2-0.2307U-67.4860$
$\text{H}_2\text{O}+\text{H}_2(\text{aq})$	$-0.5712U^2+1.0154U-46.9802$	$-0.3999U^2-0.4274U-74.7021$
$\text{H}_2\text{O}+^*$	$-0.5859U^2+0.7394U-52.8372$	$-0.4755U^2-0.4274U-74.7021$
TS_3	$-0.3446U^2+1.7242U-51.2028$	$-0.4470U^2+0.4567U-68.9926$
OH^-+H^*	$-0.4213U^2+1.9103U-51.4705$	$-0.4857U^2+1.3675U-72.6976$
$\text{H}_2\text{O}+\text{H}^*$	$-0.4038U^2+1.0148U-56.0726$	$-0.4195U^2+0.5478U-77.3312$
TS_4	$-0.4933U^2+1.4653U-54.6060$	$-0.3704U^2+1.1360U-76.2758$
$\text{OH}^-+\text{H}_2(\text{aq})$	$-0.5658U^2+1.8273U-54.6060$	$-0.4462U^2+1.3697U-77.3172$
2H^*	$-0.3919U^2+0.9579U-6.9416$	$-0.4156U^2+0.5864U-27.9428$
TS_5	$-0.4404U^2+0.8689U-6.2944$	$-0.4914U^2+0.5010U-27.5044$
$\text{H}_2(\text{aq})$	$-0.4096U^2+1.0554U-7.6682$	$-0.4247U^2+0.5716U-29.7805$

Supplementary Table 6. Calculated energy barriers and reaction free energies as a function of the electrode potential U on the Au(111) surface with θ_{*H} of 0 ML.

Reaction	$G_a(\text{eV})$	$\Delta G(U)$
$\text{H}_3\text{O}^+ + * + \text{e}^- \rightarrow \text{H}^* + \text{H}_2\text{O}$	$0.1045U^2 + 0.6964U + 0.4246$	$0.0007U^2 + 0.9868U + 0.0551$
$\text{H}_3\text{O}^+ + \text{H}^* + \text{e}^- \rightarrow \text{H}_2(\text{aq}) + \text{H}_2\text{O} + *$	$0.02059U^2 + 0.2312U + 0.7954$	$-0.1590U^2 + 0.8735U - 0.6713$
$\text{H}_2\text{O} + * + \text{e}^- \rightarrow \text{H}^* + \text{OH}^-$	$0.2413U^2 + 0.9848U + 1.6345$	$0.1646U^2 + 1.1709U + 1.3668$
$\text{H}_2\text{O} + \text{H}^* + \text{e}^- \rightarrow \text{H}_2(\text{aq}) + \text{OH}^- + *$	$-0.0895U^2 + 0.4505U + 1.4667$	$-0.1620U^2 + 0.8125U + 0.7356$
$\text{H}^* + \text{H}^* \rightarrow \text{H}_2(\text{aq}) + 2*$	$-0.0485U^2 - 0.0890U + 0.6471$	$-0.0177U^2 + 0.0975U - 0.7266$

Supplementary Table 7. Calculated energy barriers and reaction free energies as a function of the electrode potential U on Au(111) surface with θ_{*H} of 7/9 ML.

Reaction	$G_a(\text{eV})$	$\Delta G(U)$
$\text{H}_3\text{O}^+ + * + \text{e}^- \rightarrow \text{H}^* + \text{H}_2\text{O}$	$0.0591U^2 + 0.5438U + 0.8846$	$0.0118U^2 + 0.8466U + 0.6246$
$\text{H}_3\text{O}^+ + \text{H}^* + \text{e}^- \rightarrow \text{H}_2(\text{aq}) + \text{H}_2\text{O} + *$	$-0.1901U^2 + 0.1308U + 0.2508$	$-0.0391U^2 + 0.8181U - 1.2558$
$\text{H}_2\text{O} + * + \text{e}^- \rightarrow \text{H}^* + \text{OH}^-$	$-0.0471U^2 + 0.7342U + 2.2125$	$-0.0858U^2 + 0.9401U + 2.0044$
$\text{H}_2\text{O} + \text{H}^* + \text{e}^- \rightarrow \text{H}_2(\text{aq}) + \text{OH}^- + *$	$0.0491U^2 + 0.5883U + 1.0554$	$-0.0268U^2 + 0.8219U + 0.0141$
$\text{H}^* + \text{H}^* \rightarrow \text{H}_2(\text{aq}) + 2*$	$-0.0758U^2 - 0.0853U + 0.4384$	$-0.0091U^2 - 0.0148U - 1.8377$

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